

Danny Kim

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Skills

Languages: Python, TypeScript, SQL, Java

Frameworks/Libraries: Next.js, Prisma, Tailwind CSS, PyTorch, Tensorflow, MatLab

Experience

Summer Program, SSP

Jun 2026 – July 2026

- Selected to participate in an intensive five-week residential astrophysics research program focused on determining the orbit of a near-Earth asteroid.
- Developing a Python implementation to reduce observational data and apply the Method of Gauss, calculus, coordinate transformations, and orbital mechanics to calculate the asteroid's position, velocity, and six orbital elements.

Researcher, Independent

Jan 2026 – Now

- Designed and implemented a Python-based adaptive gating pipeline that analyzed Mem0's top-five retrieved records for revision markers and conflicting values, routed each query to either top-1 or top-3 context, and benchmarked the results against fixed and randomized baselines across 311 LongMemEval cases.
- Evaluated the method on 311 LongMemEval cases using Mem0, Qwen2.5-7B, Hugging Face, vLLM, Qdrant, and Python-based experimental pipelines, comparing adaptive routing against fixed top-k and randomized baselines.
- Reduced total AI reader-token usage by 37.0% and retrieved-context tokens by 53.3% while preserving 61 of 62 baseline correct answers, improving token cost per correct answer from 2,444 to 1,565.

Projects

Scioly

scioly-ten.vercel.app

- Built a full-stack operations platform with Next.js, Prisma, and role-based admin workflows, replacing spreadsheet-based processes for applications, dues, forms, and event assignments.
- Designed maintainable data models and dashboards for recurring club workflows, reducing manual coordination and improving reliability for student leadership operations.

Huracan

github.com/dantewins/huracan

- Developed an AI-assisted disaster response application that transformed messy user-submitted information into structured damage reports and actionable guidance.
- Orchestrated end-to-end workflows for intake, analysis, and response delivery, emphasizing practical usability under ambiguous real-world conditions rather than static demo behavior.

Swordle

swordle-mauve.vercel.app

- Created a SAT vocabulary game in Next.js, TypeScript, Tailwind CSS, and Supabase with solo play, login, saved progress, and win/loss streak tracking
- Implemented real-time multiplayer with matchmaking and presence detection using Supabase Realtime, extending the project beyond a static single-player game into a live competitive experience.

Emulator controller

github.com/dantewins/mumu-remote-control

- Built a workflow automation system in Node.js that orchestrated emulator control, command execution, monitoring, and recovery across Android devices through a Discord interface.
- Integrated ADB-based control, presence polling, webhook alerts, and persistent configuration into a single tool, emphasizing reliable long-running automation under real-world failure conditions.

Education

Pembroke Pines Charter High School — Junior

2023 - Present

- GPA: 3.92 UW, 5.55/6.0 W; Rank: 1/443; AP scores of 5 in Calculus AB, Biology, Computer Science A, Physics 1, and Psychology; current coursework: AP Calculus BC, AP Chemistry, AP Physics 2, AP Physics C: E&M.
- Honors & additional coursework — MCB2010 Microbiology (A); MAD2104 Discrete Mathematics (A); HOSA: 2x state qualifier, 1st and 2nd at Regionals; Science Olympiad: 1st in Circuit Lab (2026 USF Invitational), 6th in Ecology (2025 Regionals); hackathons: Shipwrecked, QuHacks, PlutoHacks, and MDC Magic.